

POST TENSIONING FORMAT FOR PSC GIRDER

General information:

Name & Location of Bridge: Construction of 301.00m Long PSC Girder Bridge Over the River Kartoa at Ch-4500 on Chatra GC Gilabari Ghat Via Nischintabati Primary School Road, Road ID - 18576-0344			District: Rangpur	Upazila: Pirganj
Span/Girder Ref: Span-1, Girder-3	Girder Length (m): 43m		Cable Ref. to be stressed: 01 of 04	
Girder Casting Date: 22-06-2026	Stressing Stage of Cable: 02		Jacking End Ref: End 1: 705017	End 2: 705016
Date of Tensioning: 09-07-2026				

Design Information:

Dia of Strand (mm): 15.24mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decortate	UTS of Strand (N/mm ²): 1860	Mt. of Elasticity, ϵ (mm/m): 10/1000
Area of Strand (mm ²): 140	Area of Cable, A (mm ²): 2664	Design Jacking Force: 3750 KN		
Design Elongation Each end (mm): 12 mm	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm): 650	Corrected Elongation for grip length d (mm): 146.25 mm	
Design Cable Slip (mm): 6mm	Design Cove. Sl. during Tensioning (N/mm ²): 30 MPa	Actual Cove. Sl. at the time of Tensioning (N/mm ²): 30 MPa		

Stressing & Jack Information:

Pump Model No. P1: 710567 P2: 716570	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m ²): TA1: TA2:	Block Stop Ram Area (m ²): BA1: BA2:
Actual Area of Cable, A1 (mm ²): 2707.50	Modulus of Elasticity, E1 (MPa): 200100	Corrected Elongation (mm) for Actual A1 & E1: $\Delta(A \cdot E / A1 \cdot E1)$: 141.32 mm	
Jack Press. With Jack loss (BAR) Kg/cm ² : J1: P/(TA1*efficiency): J2: P/(TA2*efficiency):	Calibrated Jack Press. (Kg/cm ²): C11: 440.07 C12: 449.50	Initial Jack Press. (Kg/cm ²): K11: 37.22 K12: 35.17	Initial Marking: IR11: 23 IR12: 23
Actual calculated Elongation for Grip length, Δl (mm): 4.19mm	Gross Slip of Cable (mm): [Final Elong. - Net Elong.]	Net Slip at Jack end after lock off (mm): (Gross Slip - Calcd. Elong. Δl for grip length) At end 1	
Blocking Pressure Kg/cm ² : E1: Auto Block E2: Auto Block	At end 1: 156-148=8	At end 2: 153-149=4	At end 1: 3-4.14= -1.14

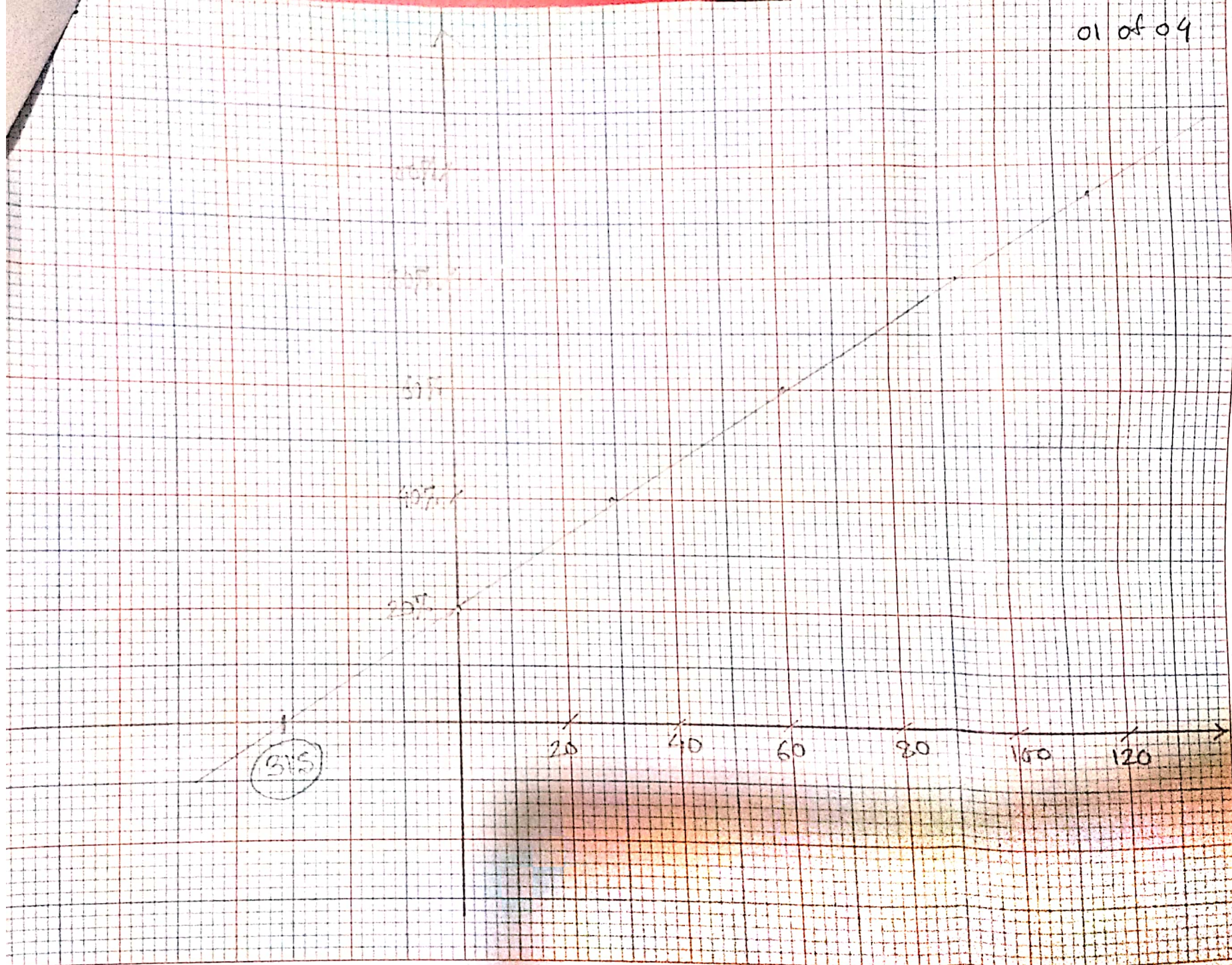
Record of Stressing & Elongation:

Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Correction Factor For ICJ	Final/ Total Elongation (mm)	Average Elongation At each End.	Remarks (Average Slip at each end)		
	Col(1)		Col(2)		Col(3)		Col(4)-(Col 3-IRI)							Col(5)	Col(6)-(4+5)
	1	2			1	2	1	2	Avg					1	2
	KN	TON	Kg/cm2	Kg/cm2	*	*	*	*	*		*	*	386/486		
20	750	76.48	87.22	85.87	39	43	*	*	*				2		
40	1500	152.96	175.43	175.53	68	72	29	29	29				4.36		
60	2250	229.44	263.65	265.19	101	104	60	61	60.5				2.6		
80	3000	305.91	351.86	354.85	132	135	93	92	92.5						
95	3562.5	363.27	418.01	422.09	156	158	117	115	116	3150		147.5	OK		
98	3675	374.75	431.25	435.09											
100	3750	382.39	440.07	444.50											
102	3825	390.04	448.89	453.97											
105	3937.5	401.51	462.12	466.92											
Lock off					148	149									

Client's Representative

Consultant's Representative

Contractor's Representative



POST TENSIONING FORMAT FOR PSC GIRDER

General Information:

Name & Location of Bridge: Construction Of 301.00m Long PSC Girder Bridge Over the River Kartoa at Ch 4500 on the left side of Ghabari Ghat Via Nischintobal Primary School Road, Road ID : 185764034			District: Rangpur	Upazila: Pirganj
Span/Girder Ref: Span-1, Girder-3	Girder Length (m): 43m	Cable Ref. to be Stressed 02 of 04		
Cable Casting Date: 22.08.2026	Stressing Stage of Cable : 02		Jacking End Ref: L709047	End Ref: L709046
Date of Tensioning: 09.07.2026				

Design Information:

Plant Strand (mm): 15.34mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decorate	UTS of Strand (N/mm ²): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm ²): 130	Area of Cable, A (mm ²): 2560	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm): 146	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm): 600	Corrected Elongation for Grip Length 0 (mm):	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm ²): 36 MPa	Actual Conc. St at the time of Tensioning (N/mm ²): 38.06 MPa		

Stressing & Jack Information:

Pump Model No: P1: 710567 P2: 710570	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m ²): TA1: TA2:	Blocking Ram Area (Cm ²): BA1: BA2:
Actual Area of Cable, A1 (mm ²): 2707.50	Modulus of Elasticity, E1 (MPa): 200-100	Corrected Elongation (mm) for Actual A1 & E1: $\delta = (A \cdot E / A1 \cdot E1): 145.18 \text{ mm}$	
Jack Press. With Jack loss (B&B) Kg/cm ² : J1: P1/A1 efficiency: J2: P2/A2 efficiency:	Calibrated Jack press. (Kg/cm ²): E1: 440.07 E2: 444.50	Initial Jack Press. (kg/cm ²): ICJ1: 87.22 ICJ2: 85.87	Initial Marking: IRJ1: 41 IRJ2: 44
Actual calculated Elongation for Grip Length, 01 (mm): 146mm	Gross Slip of Cable (mm): (Final Elong. - Net Elong.)	Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. 01 for Grip length)	
Blocking Pressure Kg/cm ² : E1: Auto Block E2: Auto Block	At end 1 160-152=8	At end 2 161-153=8	At end 1 8-414=3.86
			At end 2 8-414=3.86

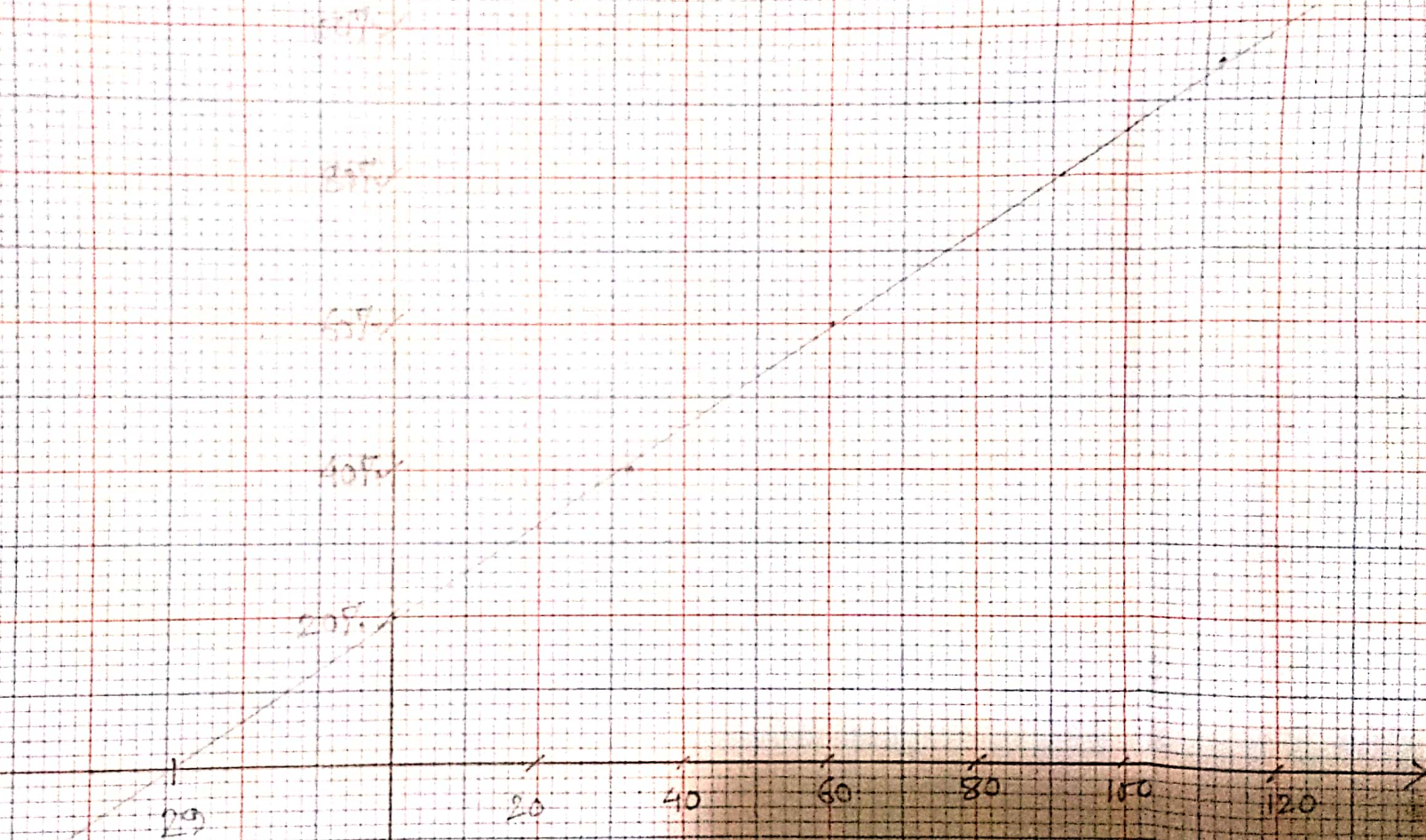
Record of Stressing & Elongation:

Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Correction Factor For ICJ	Final/ Total Elongation. (mm)	Average Elongation At each End.	Remarks (Average Slip at each end)			
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)							Col (5)	Col(6)=(4+5)	
	1	2			1	2	1	2	Avg.						1	2
	KN	TON	Kg/cm2	Kg/cm2	-	-	-	-	-		-	-				
20	750	76.48	87.22	85.87	41	44	-	-	-				3.86+3.86			
40	1500	152.96	175.43	175.53	74	76	33	32	32.5				2			
60	2250	229.44	263.65	265.19	102	104	61	60	60.5				=3.86			
80	3000	305.91	351.86	354.85	133	135	92	91	91.50				6			
95	3562.5	363.27	418.01	422.09	154	155	113	111	112				OK			
98	3675	374.75	431.25	435.09	160	161	119	117	118	29		147				
100	3750	382.39	440.01	444.50												
102	3825	390.04	448.83	453.47												
105	3937.5	401.51	462.12	466.92												
Lock off					152	153										

Client's Representative

Consultant's Representative

Contractor's Representative



POST TENSIONING FORMAT FOR PSC GIRDER

General Information:

Name & Location of Bridge: Construction Of 301.00m Long PSC Girder Bridge Over the River Kartoa at Ch 4500 on Chatra GC Gilabari Ghat Via Nischintobati Primary School Road, Road ID : 185764034				District: Rangpur	Upazila: Pirganj
Span/Girder Ref: Span-1, Girder-3	Girder Length (m): 43m		Cable Ref. to be Stressed 03 of 04		
Girder Casting Date: 22.06.2026	Stressing Stage of Cable : 02		Jacking end Ref:	End 1: L709047	End 2: L709046
Date of Tensioning: 09.07.2026					

Design Information:

Dia of Strand (mm): 15.74mm	No. of Strand Cable: 19 nos	Anchorage Brand: Decomate	UTS of Strand (N/mm ²): 1860	M. of Elasticity, E (MPa): 197000
Area of Strand (mm ²): 140	Area of Cable, A (mm ²): 2660	Design Jacking Force P: 3750 Kn		
Design Elongation Each end (mm): 150	Design Elongation for Gripping Length (mm): N/A	Actual Gripping Length (mm): 600	Corrected Elongation for Grip Length d(mm): 154.29 4.29 mm	
Design Cable Slip (mm): 6mm	Design Conc. St. during Tensioning (N/mm ²): 36 MPa		Actual Conc. St at the time of Tensioning (N/mm ²): 38.06 MPa	

Stressing & Jack Information:

Pump Model No. P1: P2:	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m2): TA1: TA2:	Blocking Ram Area (Cm2): BA1: BA2:
Actual Area of Cable, A1 (mm2) : 2702.50	Modulus of Elasticity, E1(MPA): 200400	Corrected Elongation (mm) for Actual A1 & E1 : $\delta^*(A^*E/A1^*E1)$: 149.04 mm	
Jack Pres. With Jack loss (BAR) Kg/cm2, H: P/(TA1*efficiency): J2: P/(TA2*efficiency):	Calibrated Jack press.(Kg/cm2) C1: 440.07 C2: 444.50	Initial Jack Press.(Kg/cm2) IC1: 87.22 IC2: 85.87	Initial Marking: IRJ1: 37 IRJ2: 40
Actual calculated Elongation for Grip Length, $\delta 1$ (mm): 4.14mm	Gross Slip of Cable (mm): (Final Elong. - Net Elong.)		Net Slip at Jack end after lock-off (mm): (Gross Slip - Calcul. Elong. $\delta 1$ for Grip length)
Blocking Pressure Kg/cm2): E1: Auto Block E2: Auto Block	At end 1	At end 2	At end 1 At end 2
	155-146=9	156-147=9	9-4.14=5.86 9-4.14=5.86

Record of Stressing & Elongation:

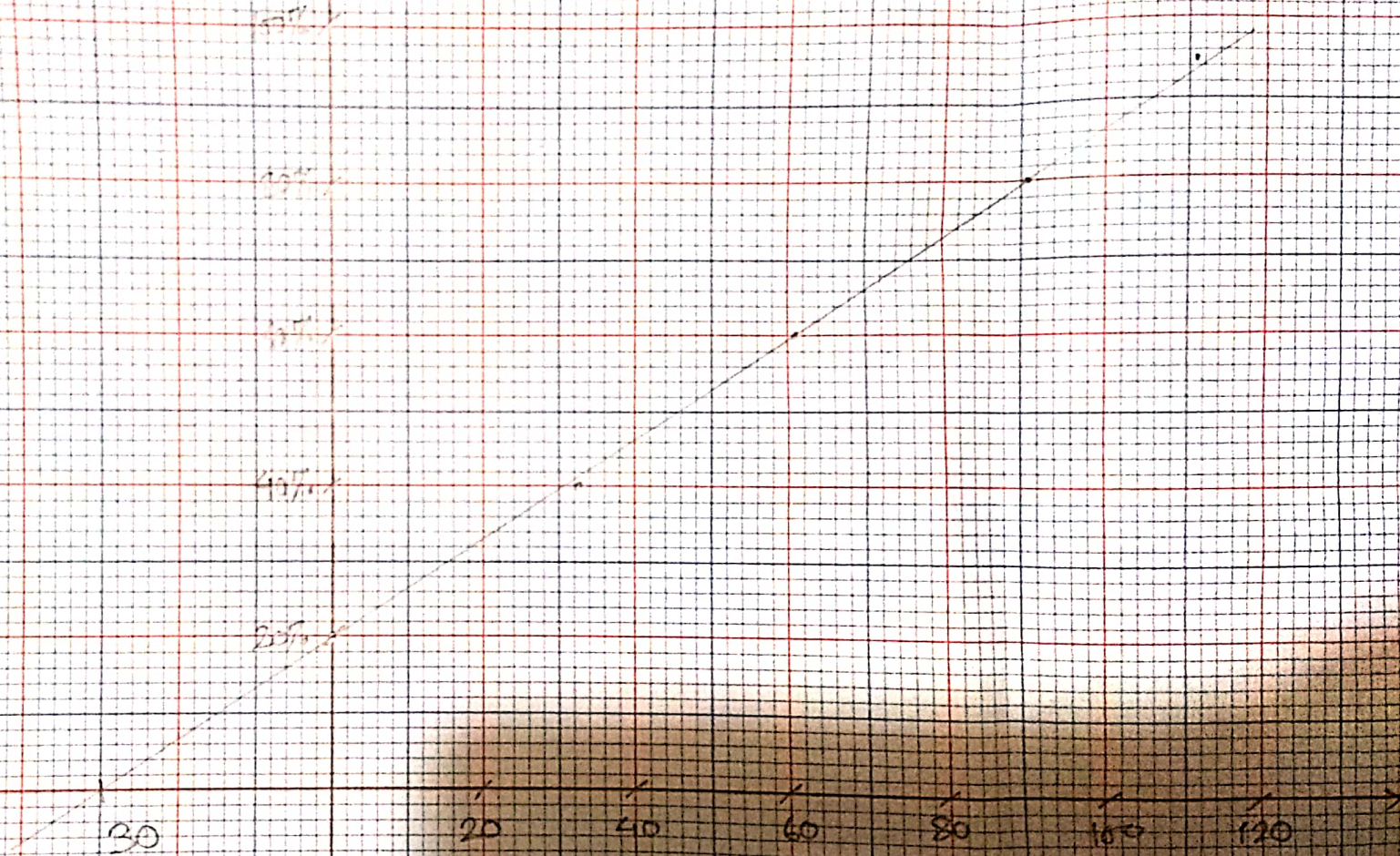
Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm ²		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Correction Factor For ICJ	Final/ Total Elongation. (mm)		Average Elongation At each End.	Remarks (Average Slip at each end)
	Col(1)	Col(2)	Col(3)	Col(4)	Col(5)	Col(6)	Col(7)	Col(8)	Col(9)		Col(10)	Col(11)		
	1	2	1	2	1	2	1	2	Avg.		1	2		
	KN	TON	Kg/cm ²	Kg/cm ²	.	.	.	.	.		.	.		
20	750	76.48	87.22	85.87	37	40	.	.	.		.	.		5.86+5.86
40	1500	152.96	175.43	175.53	69	73	32	33	32.5					2
60	2250	229.44	263.65	265.19	99	100	62	60	61					= 5.86
80	3000	305.91	351.86	354.85	128	130	91	90	90.5					
95	3562.5	363.27	418.01	422.09	149	150	112	110	111	30				< 6
98	3675	374.75	431.25	435.09	155	156	118	116	117				147	OK
100	3750	382.39	440.07	444.50										
102	3825	390.04	448.89	453.47										
105	3937.5	401.51	462.12	466.92										
Lock off					146	147								

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Girder Casting Date: 22-06-2026	Stressing Stage of Cable : 02		Jacking end Ref:	End 1: L709047
Date of Tensioning: 09-07-2026				End 2: L709046

Design Information:

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Pump Model No. P1: 710567 P2: 710570	Pressure gauge Model: M1: M2:	Tensioning Ram Area (m2): TA1: TA2:	Blocking Ram Area (Cm2): BA1: BA2:	
Actual Area of Cable, A1 (mm2) : 2707.50	Modulus of Elasticity, E1(MPA): 200400	Corrected Elongation (mm) for Actual A1 & E1 : $\delta \cdot (A \cdot E / A1 \cdot E1)$: 149.04 mm		
Jack Pres. With Jack loss (BAR) Kg/cm2. J1: P/(TA1*efficiency): J2: P/(TA2*efficiency):	Calibrated Jack press.(Kg/cm2) CJ1: 440.07 CJ2: 444.50	Initial Jack Press.(Kg/cm2) ICJ1: 87.22 ICJ2: 85.87	Initial Marking: IRJ1: 43 IRJ2: 46	
Actual calculated Elongation for Grip Length, $\delta 1$ (mm): 414mm	Gross Slip of Cable (mm): (Final Elong.- Net Elong.)		Net Slip at Jack end after lock-off (mm): (Gross Slip – Calcul. Elong. $\delta 1$ for Grip length)	
Blocking Pressure Kg/Cm2): E1: Auto Block E2: Auto Block	At end 1	At end 2	At end 1	At end 2
	161-153=8	8-414=386	165-156=9	9-414=486

Record of Stressing & Elongation:

Avg. % of Design Load	Actual Applied Pressure		Calculated Gauge Pressure Kg/cm2		Reading for Elongation (mm)		Measured Elongation At both Jacking end (mm)			Correction Factor For ICJ	Final/ Total Elongation. (mm)		Average Elongation At each End.	Remarks (Average Slip at each end)	
	Col(1)		Col(2)		Col(3)		Col(4)=(Col 3-IRJ)				Col(5)	Col(6)=(4+5)			
	1	2			1	2	1	2	Avg.			1			2
	KN	TON	Kg/cm2	Kg/cm2	-	-	-	-	-		-	-			
20	750	76.48	87.22	85.87	43	46	-	-	-					386+486	
40	1500	152.96	175.43	175.53	68	69	25	23	24					2	
60	2250	229.44	263.65	265.19	105	106	62	60	61					=2.36	
80	3000	305.91	351.86	354.85	136	133	93	93	93					6	
95	3562.5	363.27	418.01	422.09	161	165	118	119	118.5	325			151	OK	
98	3675	374.75	431.25	435.09											
100	3750	382.39	440.07	444.5											
102	3825	390.04	448.89	453.47											
105	3937.5	401.51	462.12	466.92											
Lock-off					153	156									

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